

# Radwan Al-Shawesh

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City of Sana'a, Yemen

## SUMMARY

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Computer science instructor with a strong foundation in mathematics, software engineering, and artificial intelligence. Research interests include software engineering methodologies, formal modeling, and the mathematical foundations of computing, with a growing focus on intelligent systems. Currently preparing for PhD studies, aiming to pursue rigorous, theory-driven research with practical impact.

## EDUCATION

- **East China Normal University**   
*M.S.E Software Engineering* Sep. 2018 - July 2021  
Shanghai, China
  - **Specialization:** Artificial Intelligence, Deep learning, Medical Big Data, and Computer Vision.
  - **MSc Thesis:** Histopathological Colorectal Cancer Image Classification based on Convolutional Neural Network
  - **Supervisor:** Prof. Chen YiXiang
  - **GPA:** 4.21/5.00
  - **Awards:** Shanghai Government Scholarship Type A
  - **Relevant Coursework:** Computer Architecture, Algorithm Design and Complexity Analysis, Fundamentals of Artificial Intelligence, Data Visualization, Theoretical Foundation of Software, Software/Hardware Co-Design, Machine learning, Large-Scale Nonlinear Programming, Advanced Design Methodologies for Smart Systems, Seminar on Embedded Systems.
- **Nanjing University of Aeronautics & Astronautics**   
*B.Eng Aeronautical Engineering* Sep. 2012 - July 2016  
Nanjing, China
  - **Specialization:** Aircraft Control and Information Engineering
  - **GPA:** 3.10/5.00
  - **Awards:** CSC Scholarship Type A, Top 100 Outstanding Young Students
  - **Relevant Coursework:** Linear Algebra, Algebra & Trigonometry, Introduction to Computer Science, Computer Programming, Calculus 1,2 & 3, Electrical Engineering & Electronic Technique 1 & 2, Physics 1 & 2, Information Retrieval & Utilization, Numerical Analysis, Discrete Mathematics, Mathematical Equations, Software Engineering.

## RESEARCH INTERESTS

- Machine Learning and Deep Learning
- Natural Language Processing (NLP)
- AI Applications in Healthcare
- Computer Architecture and Software System Design
- Computer Vision
- Reinforcement Learning (RL)
- Autonomous Systems
- Algorithms and Computer Science Theory

## RESEARCH EXPERIENCE

- **Enhancing Histopathological CRC Image Classification by using CNN** Sep. 2020 - March 2021  
*Tools: [Python, TensorFlow, Keras, ResNet-50, Transfer Learning, Fine-Tuning, Macenko Method]* 
  - **Institution:** East China Normal University.
  - **Objective:** Investigated convolutional neural network (CNN) techniques for improving colorectal cancer (CRC) histopathological image classification.
  - **Methods:** Adapted the ResNet-50 architecture using transfer learning and fine-tuning techniques. Leveraged pretraining on the ImageNet dataset, froze parameters in early layers, and retrained the dense and final layers to optimize feature extraction from CRC histopathological images in the NCT dataset.
  - **Outcomes:** Achieved a 97.7% validation accuracy, demonstrating the model's potential for clinical applications in cancer diagnostics.

## PATENTS AND PUBLICATIONS

### Online

- [1] R. Al.Shawesh and Y. X. Chen. "Enhancing histopathological colorectal cancer image classification by using convolutional neural network." [Online]. Available: <https://doi.org/10.1101/2021.03.17.21253390>.

Thesis

[2] R. Al.Shawesh, “Histopathological colorectal cancer image classification based on convolutional neural network and tools development,” Master’s thesis, East China Normal University (ECNU), Shanghai, China, 2021. Accessed: Dec. 30, 2024. [Online]. Available: <https://xueshu.baidu.com/usercenter/paper/show?paperid=1v1y0e90f55404w0js3p00w0p5214691>.

TEACHING EXPERIENCE

• Courses I have taught:

Teaching experience includes 1 year at the University of Science & Technology (UST).

| Course Title                     | Program | Semesters Taught | Mod of Teaching |
|----------------------------------|---------|------------------|-----------------|
| Computer Fundamentals            | BSc     | 1                | Lectures        |
| Introduction to Computer Science | BSc     | 1                | Lectures        |
| Software Engineering             | BSc     | 1                | Lectures        |

• Courses I am qualified and prepared to teach:

| Course Title                                       |
|----------------------------------------------------|
| Single Variable Calculus                           |
| Computer Programming (C/C++, Java, Python)         |
| Data Structures                                    |
| Discrete Mathematics 1 (Set Theory, Logic, Proofs) |
| Introduction to Information Science & Technology   |

PROJECTS

- **AI Pathologist: [Image Classification Python Applicaion for Lab Doctors]**

Tools: [Python, Qt for Python]

March 2021 - April 2021

- Developed a software program to utilize my own CNN network.
  - Designed the software to operate efficiently on both CPU and GPU, without requiring high-end hardware or third-party software dependencies.
  - Optimized the user interface for simplicity, allowing pathologists and lab doctors to upload, classify, and analyze images efficiently.
- **Computer Science & Technology Self Study Platform**

Tools: [Python, Sphinx, GitHub Pages, reStructuredText, Git, GitHub Actions(CI/CD)]

May 2021 - Present

- Developed a comprehensive self-study guide for Computer Science based on curricula from renowned universities like Tsinghua, MIT, and Harvard.
  - Utilized Sphinx and reStructuredText to generate clean, readable documentation, deployed to GitHub Pages using Continuous Deployment with GitHub Actions.
  - Created a structured learning roadmap, including suggested books, online resources, courses, and exams, designed to guide learners through the essential topics in computer science.
  - Integrated practical resources like the REKCARC-TSC-UHT open-source project on GitHub to enrich the guide with real-world examples and tools.
- **Data Structure Visualization using Graphviz**

Tools: [C, Graphviz]

Sep. 2022 - Present

- Developed a tool for visualizing data structures like Binary Search Trees (BST) and Threaded Binary Trees using the Graphviz library.
  - Provided interactive visualization for tree structures and operations, helping users understand internal relationships.
  - Included future support for AVL Trees and Red-Black Trees for extended functionality.

EMPLOYMENT HISTORY

Full-Time Work

- **Foshan Safe Freight CO.,LTD | Safe CareMed**

Software Engineer

July 2021 - June 2023

Foshan, China

- Customized and developed full ERPNext modules using Python and Frappe Bench, optimizing internal workflows and automating key business processes, boosting workflow efficiency by 50%.
  - Maintained Linux servers with 99.9% uptime and strengthened security.
  - Built responsive UIs with JavaScript, jQuery, HTML, and Bootstrap 5, improving usability by 25%.

- Developed and managed multiple WordPress sites, creating custom themes and plugins to enhance functionality and improve SEO ranking.
- Collaborated with the marketing team to enhance visibility, refine strategies, and actively contributed to social media marketing efforts.
- Designed visuals and video edits using Photoshop, Illustrator, and Adobe Premiere for websites, applications, and social media campaigns, enhancing brand visibility and user engagement.

**Part-Time Work**

- **University of Science & Technology** [🌐] June 2024 - Present  
Lecturer Sana’a, Yemen
  - Developed and delivered engaging lectures in *Software Engineering, Data Structures & Algorithms, Introduction to computer science, and Computer Organization* enhancing student understanding and performance.

**VOLUNTEER EXPERIENCE**

- **Volunteer Teaching** June 2015 - July 2015  
*Gansu Province - Baiying city - Hegan Junior High School* [🌐]
  - Taught science in English to junior high school students during a 15-day volunteer teaching and cultural exchange program.
  - Promoted cross-cultural understanding and improved students’ English communication skills through interactive science lessons.
  - Recognized as an NUAA Advanced Individual of Summer Social Practice for contributions to education and community development.

**HONORS AND AWARDS**

- **Shanghai Government Scholarship Type A** Sep. 2018  
*East China Normal University* [🌐]
  - Awarded this scholarship to pursue a Master’s degree in Software Engineering.
  - Successfully completed a three year master degree at ECNU.
- **China Government Scholarship Type A** Sep. 2011  
*Embassy of China in Yemen* [🌐]
  - Awarded this scholarship to pursue a bachelor degree.
  - Successfully completed a one-year preparatory course at Tongji University.
  - Successfully completed a four-year bachelor’s degree at NUAA.
- **Top 100 Students Award** Sep. 2015  
*Nanjing University of Aeronautical & Astronautical Engineering* [🌐]
  - This award was given for excellence in academic performance, recognizing me as one of the top 100 students in the university during the 2015-2016 academic year.
- **Certificate of Honor** June 2012  
*Tongji University* [🌐]
  - Won the Outstanding Academic Achievement Award, Excellent Attendance Award, and Active Participation in Various Activities Award.

**SKILLS**

- **Programming Languages:** C/C++, Python, Java, SQL
- **Data Science & Machine Learning:** Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn
- **Operating Systems:** Linux(Debian, Ubuntu, Fedora), Windows
- **Database Systems:** MySQL, PostgreSQL, MongoDB, SQLite
- **DevOps & Version Control:** Git, GitHub, CI/CD
- **Web Technologies:** HTML, CSS, Bootstrap, JavaScript, React, Node.js
- **Mathematical & Statistical Tools:** MATLAB
- **Software Engineering:** UML, Object-Oriented Design (OOD), Software Architecture, Design Patterns, Software Development Lifecycle (SDLC)
- **Teaching Skills:** Curriculum Design, Classroom Management, Assessment & Evaluation, Student Mentoring, Lecture Preparation
- **Research Skills:** Experimental Design, Data Analysis, Scientific Writing, Literature Review, Research Methodologies, Grant Writing
- **Other Tools & Technologies:** LaTeX, Jupyter Notebooks, Microsoft Office, Photoshop, Illustrator, Premiere

**ADDITIONAL INFORMATION**

- Languages:**
- **Arabic:** Native speaker.
  - **English:** Fluent in reading, writing, and listening, with average speaking skills. Completed a bachelor’s program taught in English and published academic papers in English. My primary, middle, and high school education was primarily in English.
  - **Chinese:** HSK Level 6 (highest proficiency level).

## REFERENCES

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1. **Professor Chen YiXiang (陈仪香) [🌐]**  
Professor, College of Software Engineering  
East China Normal University  
Email: yxchen@sei.ecnu.edu.cn  
Phone: +86-186-2181-5769  
*Relationship: Master's Degree Academic Advisor*
2. **Prof. (Dr) Wang Wei Ying [🌐]**  
Professor, Department of Engineering  
Nanjing University of Aeronautical & Astronautical Engineering  
Email: meewywang@nuaa.edu.cn  
Phone: +86-138-5151-4220  
*Relationship: Head Supervisor during a Design Project*
3. **Prof. (Dr) Qian Yan (钱焱) [🌐]**  
Professor, Department of Business Administration  
Nanjing University of Aeronautical & Astronautical Engineering  
Phone: +86-25-8489-6220  
*Relationship: Instructor*
4. **Dr. Li Ting (李挺) [🌐]**  
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*Relationship: Head teacher*